

Gaurav Bhatia

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Supply Chain & Demand Planning Analyst

Operations and supply chain analyst finishing an MSc in Data Science at GISMA in December 2026. Monitored inventory and demand across **700+ vendors and 500+ SKUs** at Troika, rescued a **₹2.75M export order** by moving it to air freight, and ran a business for eight years, cutting **excess-inventory cost by 10%**. Adds forecasting: Prophet at **6.07% MAPE** on weekly SKU demand. Works in Tally, Excel and Power BI. Available full-time in Berlin from **December 2026**.

Skills

Supply chain & operations	Inventory and demand monitoring, demand forecasting (ARIMA, Prophet, XGBoost), stock-level and reorder analysis, supplier and vendor management, export logistics and freight-mode decisions, Tally ERP
Charts & reporting	Power BI / DAX, Plotly, Streamlit dashboards, automated Markdown/HTML KPI reports, Excel, Tally
Programming	Python (pandas, NumPy, statsmodels, scikit-learn, XGBoost), SQL (CTEs, window functions), DuckDB, MySQL, SQLite
Methods	Demand forecasting (ARIMA, Prophet, XGBoost), regression and price elasticity, RFM segmentation, anomaly detection, model evaluation (MAE, RMSE, MAPE)
Languages	English (C1), German (A2, improving), Hindi (native)

Experience

Data Science Intern — Aug 2026 – Present
KnowGuard AI — *Berlin, Germany*

- Ran a **paired A/B test on AWS** comparing a fine-tuned model against the review arm: **+0.0318 AUPRC** (95% CI 0.022–0.0418), below the pre-set 0.05 practical-effect threshold

Operations & Supply Chain Analyst — Mar 2024 – Jul 2025
Troika International Pvt. Ltd. — *Panchkula, India*

- Monitored inventory and demand across **700+ vendors** and **500+ SKUs** in Tally and Excel
- Worked directly with operations and supply-chain stakeholders to turn recurring questions into **standing reports**
- Rescued a stalled **₹2.75M export order** (500 units to Lithuania) by moving the shipment to **air freight** to hold the delivery window and renegotiating terms with the client
- Was handed that order **end to end** and ran every part: sourced and booked it at the manufacturer in person, costed it at ₹5,500/unit, arranged freight and carried the client through to delivery
- Built **Power BI dashboards across 3 divisions**, turning operational data into clear views for non-technical stakeholders

Business Manager — Nov 2015 – Jan 2024
Bhatia Traders — *India*

- Ran a product business end to end for **8 years**, owning purchasing, pricing, stock and supplier terms across **20+ suppliers**
- Analysis-led changes cut **excess-inventory cost by 10%** and identified a ~**15% stockout-reduction** potential
- Ran three ventures alongside it: a **home-tutoring marketplace**, **Shopify e-commerce stores** and a restaurant
- Demand patterns from this business became the basis of the **Demand Forecasting Engine** project

Education

GISMA University of Applied Sciences — Potsdam, Germany
MSc Data Science, AI & Digital Business — *Sep 2025 – Dec 2026 (expected)*

- Thesis: the US–Iran conflict's impact on the German electricity market

University College of Engineering, Patiala — *B.Tech, Civil Engineering, 2011 – 2015*

Selected Projects

Demand Forecasting Engine — *Python, ARIMA, Prophet, XGBoost, Streamlit* — Source

- Compared ARIMA, Prophet and XGBoost on 5 years of weekly sales for 12 SKUs on a 12-week hold-out; **Prophet best at 6.07% MAPE** vs. 15.49% for ARIMA
- Shipped as a Streamlit app where a planner picks SKU, model and horizon and sees MAE/RMSE/MAPE live

Bhatia Traders Sales Analysis — *Python, pandas, Matplotlib* — Source

- Analysed Bhatia Traders sales with moving averages and a margin ranking by product, and sized a ~**15% stockout-reduction** potential
- The demand patterns it surfaced became the basis of the Demand Forecasting Engine

Inventory Alert System — *MySQL (CASE, JOINs, UPDATE)* — Source

- Built MySQL queries that raise automated **low-stock alerts at a 20% threshold** with reorder recommendations for a sauces and condiments business

Dynamic Pricing & Segmentation *Python, statsmodels, XGBoost, Plotly* Source

- Estimated **price elasticity per category** with log-log OLS on 15,000 SKUs in 8 categories (e.g. Fashion −2.1, Food & Beverages −0.5)
- Segmented customers with **RFM scoring**; 6 business rules + XGBoost repriced **55.7% of the catalogue** (8,352 SKUs), and an XGBoost price recommender reached **2.14% MAPE**

BI Data-Quality Pipeline *Python, pandas, NumPy* Source

- Automated **45 validation checks** (missing values, duplicates, referential integrity, outliers, row counts) across a 7-table star schema; **41/45 passed (91.1%)**
- Traced two silent data losses (614 payment rows, 551 review rows) to root cause and documented fixes in an auto-generated report

MSc Thesis: Conflict Shocks in the German Power Market *Python, Chronos, XGBoost, Granger causality* Source

- Built a reproducible pipeline answering **5 research questions** on how the 2026 US–Iran conflict moved German day-ahead electricity prices, benchmarked against the 2022 energy shock
- Combined **event study**, **Granger causality** and ablation tests with **Chronos** and XGBoost forecast arms to identify the transmission channel

Payments BI Dashboard *Power BI, DAX, Python, star schema* Source

- Built a 3-page **Power BI** dashboard over **100K payment transactions** ($15.99M revenue) covering BNPL share, delivery performance and revenue mix with **DAX** measures

Toll Traffic Reporting and Anomaly Suite *DuckDB, SQL, Isolation Forest, APScheduler, Jinja2* Source

- Built **SQL reporting** (CTEs, window functions, RANK), **Z-score and Isolation Forest** anomaly detection, and a scheduled weekly HTML KPI report over German highway toll data

Office Energy Forecasting Dashboard *XGBoost, Optuna, SHAP, Streamlit* Source · Demo

- Tuned XGBoost with **50 Optuna trials** to predict office energy use to **MAE 22.98 Wh** and RMSE 50.96 Wh, surfaced with CO_2 savings in a live Streamlit app
- **SHAP** showed short-term momentum dominates consumption, driving out-of-hours waste alerts

Certifications

- **Introduction to Data Engineering with Microsoft Azure 1 & 2**, FutureLearn, Sep 2026
- **Introduction to Cloud Infrastructure: Describe Cloud Concepts**, Microsoft, Jul 2026
- **Data Analytics Job Simulation**, Deloitte, issued by Forage, Jul 2026